

Physics 0525: Lab 6

Bipolar Transistors

Equipment: breadboard, DMM, oscilloscope, function generator, 2N3904 npn transistor, capacitors, decade resistor box, resistors

Preliminaries:

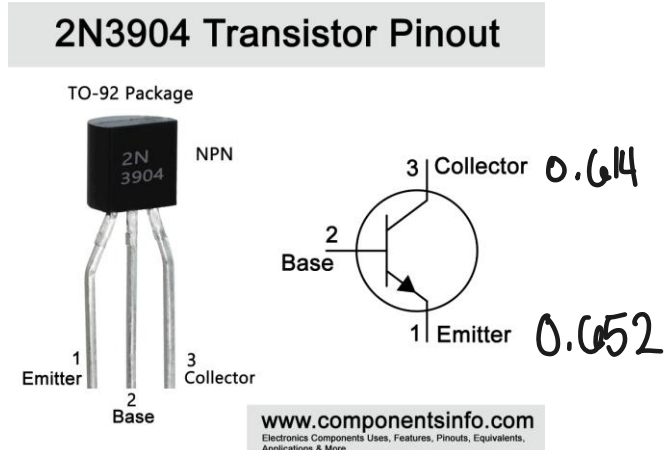
The transistor is the most important active electronic component, a device that amplifies an input signal and produces an output signal with more power (with the help of a power supply). It is a 3-terminal device available in two flavors, *npn* and *pnp*. In this lab, you will use a *npn* transistor made by sandwiching a p-type semiconductor between two n-type semiconductors with electrical contacts connected separately to each region (Figure 1).

In principle, the bipolar transistor behaves like a switch or a current amplifier. The base terminal connected to the middle region allows control of the current flow between the other two terminals (collector and emitter). The basic rule of thumb for normal operation of the *npn* transistors (for *pnp* transistors the polarities have to be reversed):

- (1) The collector must be held at higher potential (more positive) than the emitter ($V_C > V_E$).
- (2) The base-emitter and base-collector junctions behave like diodes. Normally the base-emitter diode is forward biased and the base-collector diode is reverse biased (i.e. $V_C > V_B > V_E$). Typically, a resistor connected to the base controls the current flowing into the base, I_B , by keeping the base voltage about 0.6 V above the emitter voltage ($V_B \approx V_E + 0.6V$).
- (3) The current from collector to emitter, I_C , is proportional to the base current: $I_C = \beta I_B$. The current gain (β) is in the range of 50-200 depending on the transistor. Due to conservation of charge, the total emitter current is $I_E = I_B + I_C$.

Before you start working, identify the emitter, base, and collector leads of your transistor. Wrong connections may destroy the transistor. Use the DMM diode-test function to measure the base-to-collector and base-to-emitter voltages in forward and reverse directions just like you did for the diode in the previous lab. Faulty transistors usually exhibit shorts in both directions.

Figure 1. Pin assignments of the 2N3904 npn transistor viewed from its flat side.



Name: _____

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Part 1: Emitter follower circuit operating between -15 V and $+15\text{ V}$

Assemble the circuit in Figure 2 without the blocking capacitor and the $1\text{ k}\Omega$ load resistor. Test the circuit with a sine wave from the function generator. Use both channels of the scope to display input and output at the same time.

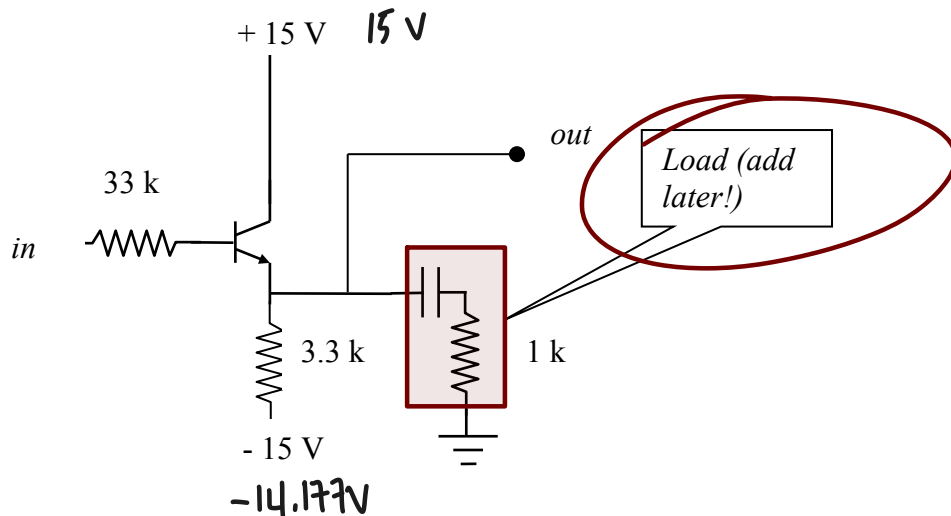
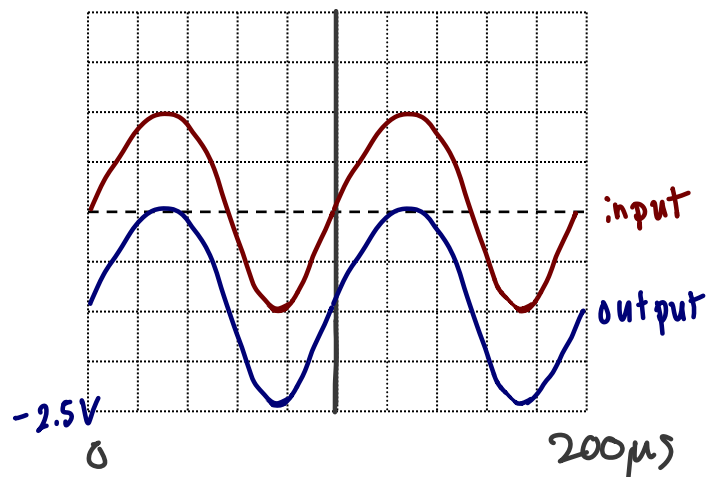


Figure 2: Emitter follower

Sketch the input and output:



Are the input and output in-phase or out of phase?

in phase

Input impedance: Without the $1\text{ k}\Omega$ load resistor, measure the peak-to-peak amplitude (V_{pp}) of the input signal before and after the $33\text{ k}\Omega$ base resistor using the oscilloscope. From the ratio of the two voltages, you can calculate the input impedance of the emitter follower circuit at the base of the transistor.

Signal amplitude before $33\text{ k}\Omega$ base resistor: 2 $[V_{pp}]$

Signal amplitude after $33\text{ k}\Omega$ base resistor: 1.88 $[V_{pp}]$

Name: _____

Date: _____

$$R_{in} = \frac{V_B}{V_A - V_B} \cdot R \Rightarrow \frac{1.88}{2 - 1.88 \text{ V}} \cdot 33 \text{ k}\Omega \Rightarrow 517,000$$

Hence: Input impedance of the follower at the base: $R_{in} = \underline{517 \text{ E } 3} \text{ } [\Omega]$

Output impedance: Now, measure the amplitude of the output signal with and without the load resistor. Use a 10 μF electrolytic capacitor (with - negative lead to emitter!) to couple the 1 k Ω load resistor. From the ratio of the two voltages, you can calculate the output impedance of the emitter follower circuit at the emitter of the transistor.

Output amplitude without load resistor: 1.88 [V_{pp}]

Output amplitude with 1 k Ω load resistor: 1.6 [V_{pp}]
0.99

$$R_{out} = \frac{V_{out}^{no L} - V_{out}^L}{\frac{V_{out}^L}{R_L}} = \frac{1.88 - 1.6}{\frac{1.6}{1 \text{ k}\Omega}} \Rightarrow \frac{0.28}{1.6}$$

Hence: Output impedance of the follower: $R_{out} = \underline{175} \text{ } [\Omega]$

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Why is the input and output impedance of the follower significant? Why is this circuit useful?

the emitter follower allows you to add an output load
without causing a voltage drop.

The input and output impedance

Part 2: Emitter follower circuit with single power supply

This circuit is similar to the previous one, but it uses only a single power supply. The base bias is established by a voltage divider. Use a non-polarized blocking capacitor (e.g. 0.33 μ F) at the input. Build the circuit and test it with sine waves of small amplitudes.

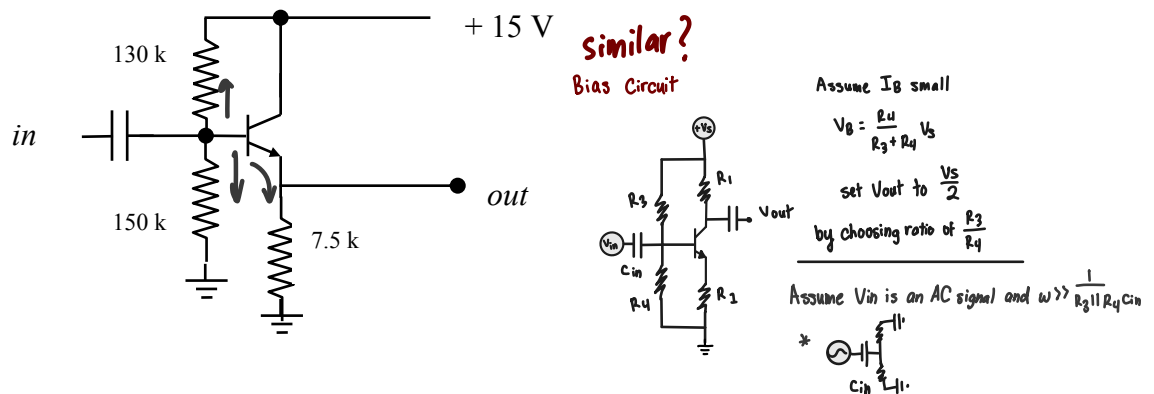


Figure 3: Single-supply follower

What is the DC offset at the base of the transistor? Why?

$$V_B = 15 \cdot \frac{150}{130+150} = 8.04V$$

Gradually increase the input amplitude until the output begins to “clip”. A good amplifier should start clipping at the positive and negative sides at the same time. Why? Increase the amplitude of the input signal until the output is clipped.

An amplifier should be biased near the middle of its output range.

This will in turn yield the cleanest signal.

What is the maximum amplitude that can be achieved without clipping?

$$\approx 7V_{pp}$$

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Date: _____

Part 3: Measuring the DC current gain of the transistor

In this part of the exercise, you measure the DC current gain β of the transistor. Use two DMMs to measure the current flow into the base and the collector at the same time (one of the ammeter in series with the collector is shown in the figure below; insert the second one in series with the base resistors). An alternative and safer option for the DMM is to measure the voltage drop across the 47 k Ω base resistor and the 1 k Ω collector resistor and calculate the currents from Ohm's law.

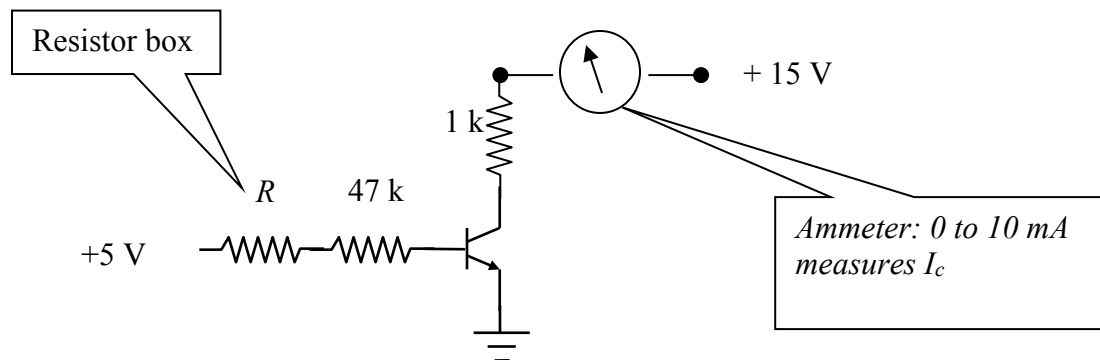


Figure 4

The base current is varied by changing the setting of the decade resistor box (works up to ~1 M Ω) or adding other resistors. A reasonable range for the base current is from 1 to 100 μ A. Try about 10 values for R in the range from 50 k Ω to 4.7 M Ω . The ratio of the collector current to the base current is β (the current gain).

Data Table:

R k Ω	I_b μ A	I_c mA	β
50	44.9	8	178.17
150	22	3.775	171.59
250	14.64	2.396	163.66
350	10.955	1.772	161.75
450	8.76	1.327	151.48
550	7.3	1.07	146.58
650	6.24	.89	142.63
750	5.47	.756	138.21
850	4.86	.656	132.74
950	4.375	.578	132.11

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Part 4: Common-emitter (inverting) amplifier

Construct the common-emitter amplifier shown in Figure 6 and test it with low-amplitude (<1 V) 1 kHz sine waves. Make sure the amplitude is low enough so that the output is not clipped.

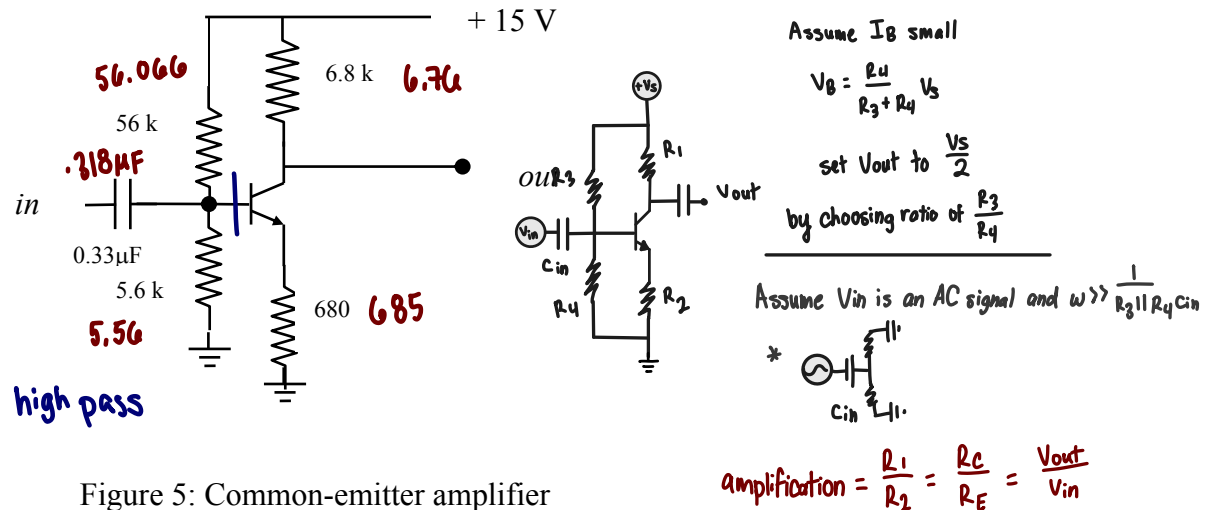


Figure 5: Common-emitter amplifier

What is the phase relation between input and output?

In phase? [~~X~~] 180° out phase? [☒]

Calculate the expected voltage gain (i.e. the ratio of the output and input amplitude).

Calculated Voltage gain: 10 V

$$|A_v| = \frac{R_C}{R_E} = \frac{6.8 \text{ k}\Omega}{680 \Omega} = 10$$

Measure the voltage gain, i.e. the ratio of peak-to-peak amplitude of the input and output on the oscilloscope.

Measured V_{in} : 9.84 mV [V_{pp}]

Measured V_{out} : 9.04 [V_{pp}]

Voltage gain: 9.147

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Measure the amplitude of the signal across the 680Ω emitter resistor.Voltage across emitter resistor: 928 [V_{pp}]

In phase? [✓] 180° out phase? []

The capacitor and resistors at the input of the circuit act like a high-pass filter. Gradually reduce the frequency of the input signal and measure the -3dB low-frequency cut-off of the circuit at the emitter. Does it agree with the expected cut-off frequency based on the capacitor and resistor values?

Low-frequency cut-off occurs at $f =$ 805.064 [Hz]When $\frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{2}}$, $f_{cut\ off}$ occurs:

$$f_c \Rightarrow \frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{2}}:$$

$$\text{the cutoff frequency} \Rightarrow f_c = \frac{1}{2\pi RC} \Rightarrow \frac{1}{2\pi \left(\frac{1}{56.064k} + \frac{1}{5.56k} \right)}$$

 $\hookrightarrow 805\text{ Hz}$

Measure the amplitude of the output at the collector with and without a blocking capacitor (10 μF , with + lead to collector) and resistive load (e.g. 1 k Ω), and determine the output impedance of the circuit as you did in Part 1.

Output impedance: $Z_{out} =$ 6.715 E3 [Ω]

$$R_{out} = \frac{V_{out}^{no\ L} - V_{out}^L}{\cancel{V_{out}^L} / R_L}$$

L: 1.16V	}	$\frac{9.04V - 1.16}{1.16 / .4886k} \Rightarrow 6715.66 \uparrow$
no L: 9.04V		

How does the output impedance compare to that of the follower in Part 1? Why is this a problem?

The impedance is far greater than that recorded in part 1.

This impedance fights the effect observed in 1 where load didn't cause V_{drop} .